Mutual Fund Analytics

Final Capstone Project Report

Submitted By:
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Technology Stack:

Python, Pandas, SQLite, SQL, Streamlit, Power BI

Bluestock Data Engineering & Analytics Internship

Summary:

This project focuses on building an end-to-end Mutual Fund Analytics platform. The solution collects, cleans, validates, stores, and analyses mutual fund data to generate business insights. The project includes ETL pipelines, database integration, exploratory data analysis, fund performance evaluation, advanced risk analytics, and interactive dashboards using Power BI and Streamlit.

Objectives:

• Build an automated mutual fund data pipeline.
• Clean and validate financial datasets.
• Store processed data in SQLite.
• Perform exploratory data analysis.
• Analyze fund performance using financial metrics.
• Calculate risk measures such as VaR and Sharpe Ratio.
• Develop investor analytics and recommendation systems.
• Create interactive dashboards for business users.

Data Sources

The Mutual Fund Analytics project uses multiple datasets to perform comprehensive financial analysis, investor behavior analysis, and risk assessment.

1. NAV History Dataset

The NAV (Net Asset Value) History dataset contains daily NAV values for multiple mutual fund schemes from 2022 to 2026. This dataset is used to analyze fund growth, calculate daily returns, compute CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR metrics.

Key Fields:

- Date
- AMFI Code
- Scheme Name
- NAV Value

2. Investor Transactions Dataset

The Investor Transactions dataset contains detailed records of investor activities, including SIP investments, lump-sum investments, and redemptions. This dataset is used for investor analytics, cohort analysis, SIP continuity analysis, demographic studies, and geographic distribution analysis.

Key Fields:

- Investor ID
- Transaction Date
- Transaction Type
- Investment Amount
- State
- Age Group
- City Tier

3. Scheme Performance Dataset

The Scheme Performance dataset contains fund-level performance information such as AUM, expense ratio, returns, ratings, and risk grades. This dataset is used to evaluate mutual fund performance and generate fund rankings and recommendations.

Key Fields:

- AMFI Code
- Fund House
- Category
- AUM (Assets Under Management)
- Expense Ratio
- 1-Year Return
- 3-Year Return
- 5-Year Return
- Risk Grade
- Morningstar Rating

4. Portfolio Holdings Dataset

The Portfolio Holdings dataset contains the allocation of fund investments across various sectors and securities. This dataset is used to evaluate diversification and concentration risk through the Herfindahl-Hirschman Index (HHI).

Key Fields:

- AMFI Code
- Sector Name
- Security Name
- Allocation Weight (%)
-

Data Quality Measures

Before analysis, all datasets underwent data validation and cleaning processes, including:

- Missing value handling
- Duplicate record removal
- Data type standardization
- AMFI code validation
- Consistency checks across datasets

These processed datasets were then loaded into SQLite and used for analytics, reporting, and dashboard development.

ETL Architecture

The Mutual Fund Analytics project follows a structured ETL (Extract, Transform, Load) architecture to ensure reliable, accurate, and scalable data processing. The ETL pipeline transforms raw mutual fund datasets into clean and analysis-ready data that supports business intelligence dashboards and advanced financial analytics.

ETL Workflow

Data Sources
↓
Data Validation
↓
Data Cleaning & Transformation
↓
SQLite Database
↓
Analytics & Risk Modelling
↓
Dashboards & Reports

1. Data Extraction (Extract)

The first stage of the pipeline involves collecting mutual fund datasets from multiple sources. These datasets include historical NAV records, investor transaction data, scheme performance metrics, and portfolio holdings information.

The extraction process loads raw CSV files into Python using Pandas and performs initial inspections to verify file integrity and structure.

Datasets Extracted:

- NAV History
- Investor Transactions
- Scheme Performance
- Portfolio Holdings

2. Data Validation

Before processing, all datasets undergo validation checks to ensure data quality and consistency.

Validation activities include:

- Verification of AMFI codes
- Missing value identification
- Duplicate record detection
- Data type verification
- Range and consistency checks

This stage ensures that inaccurate or incomplete records do not affect downstream analytics.

3. Data Cleaning and Transformation

The transformation layer standardizes and prepares data for analysis.

Key transformation activities include:

- Handling missing values
- Removing duplicate records
- Standardizing date formats
- Converting numeric fields to appropriate data types
- Creating derived metrics
- Normalizing categorical fields

The cleaned datasets are then stored in the processed data layer for further use.

4. Data Loading into SQLite

After transformation, the cleaned datasets are loaded into a SQLite database.

SQLite serves as the central storage layer for:

- Historical NAV records
- Investor transactions
- Fund performance metrics

The database structure improves query performance and enables efficient analytical operations using SQL.

5. Analytics and Risk Modelling

The processed data is then used for multiple analytical workflows, including:

Exploratory Data Analysis (EDA)

- NAV trend analysis
- SIP inflow analysis
- Investor demographic analysis

Performance Analytics

- CAGR calculations
- Sharpe Ratio
- Sortino Ratio

- Alpha and Beta estimation
- Maximum Drawdown analysis

Advanced Risk Analytics

- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling Sharpe Ratio
- Investor cohort analysis
- SIP continuity analysis
- Fund recommendation system
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6. Dashboard and Reporting Layer

The final layer presents insights through interactive dashboards and reports.
The project includes:

- Streamlit Dashboard
- Power BI Dashboard
- Analytical Reports
- Risk Analysis Reports

These dashboards provide stakeholders with a user-friendly interface to monitor fund performance, investor behavior, risk metrics, and market trends.

Benefits of the ETL Architecture

The implemented ETL architecture provides several advantages:

- Improved data quality and reliability
- Automated processing workflows
- Centralized data storage
- Efficient analytical processing
- Scalable dashboard integration
- Better decision-making through actionable insights

This architecture forms the foundation of the Mutual Fund Analytics platform and enables end-to-end data engineering and business intelligence capabilities.

Database Design

The Mutual Fund Analytics project uses SQLite as the primary database management system for storing and managing processed mutual fund data. After data validation and cleaning, the datasets are loaded into SQLite tables to enable efficient querying, reporting, and analytical processing.
The database follows a fact-table design where each table stores a specific category of information related to mutual fund operations and performance.

1. fact nav

The fact nav table stores historical Net Asset Value (NAV) records for all mutual fund schemes included in the project.
NAV represents the per-unit value of a mutual fund and serves as the foundation for return calculations and performance evaluation.

Purpose:

- Track daily NAV movements.
- Calculate daily returns.
- Compute CAGR and performance metrics.
- Support trend analysis and risk calculations.

Key Fields:

- AMFI Code
- Scheme Name
- Date
- NAV Value

Business Importance:
This table is essential for analyzing fund growth patterns and evaluating long-term investment performance.

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2. fact transactions

The fact transactions table contains investor transaction records, including SIP investments, lump-sum investments, and redemption activities.
The table captures investor behaviour and serves as the basis for investor analytics and transaction trend analysis.

Purpose:

- Analyze investment patterns.
- Track SIP continuity.
- Study investor demographics.
- Perform cohort analysis.

Key Fields:

- Investor ID
- Transaction Date
- Transaction Type
- Amount Invested
- State
- Age Group
- City Tier

Business Importance:
This table helps identify investor trends and supports decision-making related to customer engagement and retention.

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3. fact performance

The **fact performance** table stores fund-level performance information and analytical metrics.
The table includes return measures, risk grades, ratings, and fund characteristics used throughout the project.

Purpose:

- Evaluate fund performance.
- Rank mutual funds.
- Support recommendation systems.
- Compare risk-adjusted returns.

Key Fields:

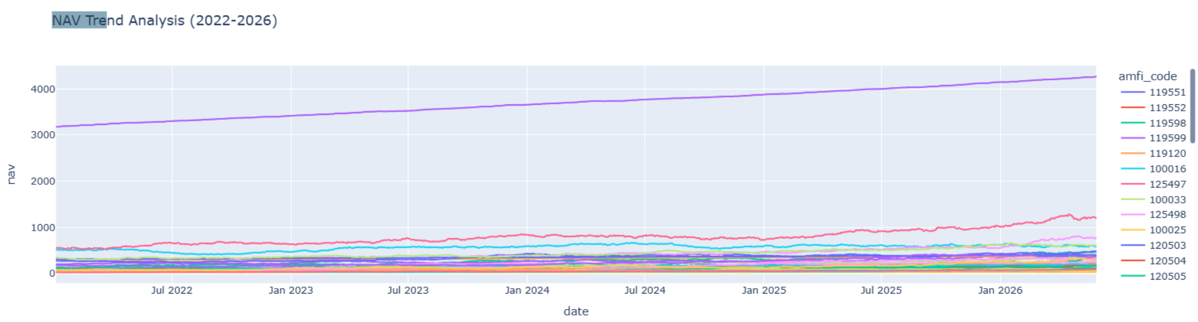
- AMFI Code
- Fund House
- Category
- AUM
- Expense Ratio
- 1-Year Return
- 3-Year Return
- 5-Year Return
- Risk Grade
- Morningstar Rating

Business Importance:
This table serves as the primary source for performance analytics and fund recommendation logic.

Database Benefits
The SQLite database provides a centralized and structured storage layer that improves analytical efficiency and supports complex SQL queries. The database enables seamless integration between data engineering workflows, analytical notebooks, and dashboard visualizations.
The use of SQLite ensures portability, ease of deployment, and efficient access to mutual fund data for reporting and business intelligence purposes.

Exploratory Data Analysis (EDA) – Part 1

NAV Trend Analysis
A time-series analysis was performed on historical NAV data from 2022 to 2026 to understand fund performance trends. Most mutual fund schemes showed steady NAV growth over the period, reflecting positive long-term performance. Temporary declines were observed during market corrections, followed by recovery during bullish phases.
Key Observations:
- Most funds demonstrated consistent NAV growth.
- Equity funds recorded stronger growth compared to conservative funds.
- Market volatility caused short-term fluctuations.
- Long-term performance remained positive across major schemes.



AUM Growth Analysis
Assets Under Management (AUM) analysis was conducted to evaluate fund popularity and investor confidence. The results showed that large asset management companies maintained higher AUM levels and attracted significant investor inflows.
Key Observations:
- Leading fund houses dominated total AUM.
- Investor participation increased over time.
- Equity-oriented funds attracted higher investments.
- Strong AUM growth reflected positive market sentiment.

Business Insight
The combination of rising NAV values and growing AUM indicates increasing investor confidence and a healthy mutual fund ecosystem. These findings provide a strong foundation for further performance and risk analysis.

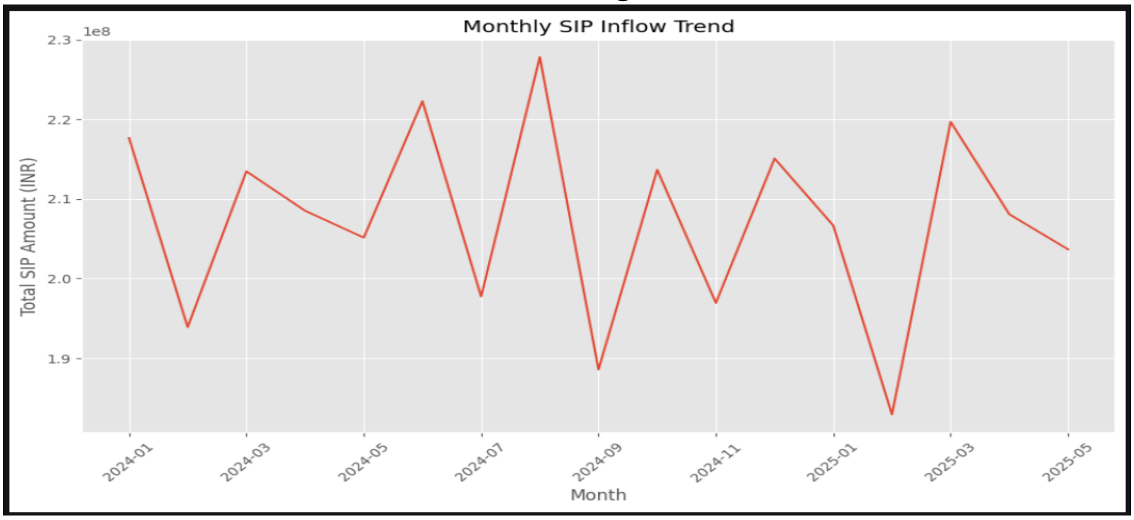
Exploratory Data Analysis (EDA) – Part 2

SIP Inflow Analysis

A monthly SIP (Systematic Investment Plan) inflow analysis was conducted to understand investor participation trends. The results showed a consistent increase in SIP contributions over the years, indicating growing awareness and adoption of disciplined investing.

Key Observations:

- SIP inflows increased steadily over the analysis period.
- Investor participation grew significantly.
- Monthly investment patterns remained consistent.
- SIP investments contributed to long-term wealth creation.



Investor Demographics Analysis
Investor demographic data was analyzed using age groups and gender distribution. The analysis revealed that young and middle-aged investors accounted for the majority of mutual fund investments.
Key Observations:

- Young and middle-aged investors formed the largest investor segment.
- SIP investments were popular across all age groups.
- Investment participation was distributed across both genders.
- Regular investing behavior was observed among long-term investors.

Business Insight
The growing SIP trend and increasing participation from younger investors indicate strong long-term growth potential for the mutual fund industry. These patterns highlight the importance of systematic investing and financial awareness among retail investors.

Exploratory Data Analysis (EDA) – Part 3

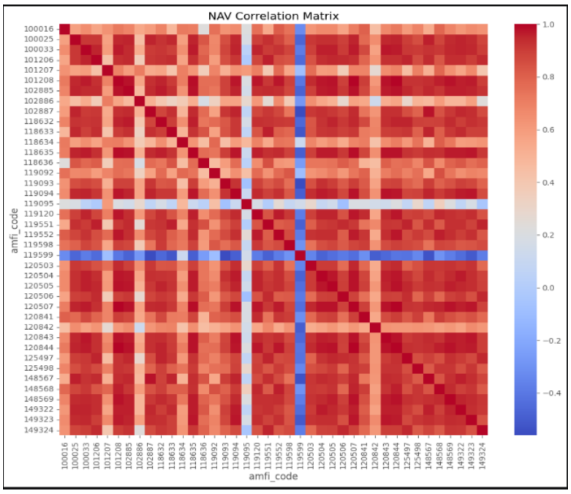
Geographic Distribution Analysis
A geographic analysis was performed to understand investment patterns across different states and city tiers. The results showed that investments were concentrated in major urban regions, while participation from smaller cities continued to grow steadily.
Key Observations:

- Metropolitan cities contributed the highest investment volumes.
- Investor participation increased across multiple states.
- Both T30 and B30 cities contributed to mutual fund growth.
- Geographic diversification reduced dependency on a single region.

Correlation Analysis
A correlation matrix was created to measure the relationship between returns of selected mutual fund schemes. The analysis helped identify funds that moved together and those that offered diversification benefits.
Key Observations:

- Funds within similar categories showed higher correlation.
- Diversified funds exhibited lower correlation with sector-specific funds.
- Correlation analysis helped identify portfolio diversification opportunities.
- Some funds demonstrated independent return behavior, reducing overall portfolio risk.

Business Insight
The geographic analysis highlighted the expanding reach of mutual fund investments across India, while the correlation analysis demonstrated the importance of diversification in managing portfolio risk and improving long-term investment performance.



Performance Analytics – CAGR & Sharpe Ratio

CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated to measure the annualized growth rate of mutual funds over different investment horizons. CAGR provides a standardized method for comparing fund performance across multiple schemes.

Key Observations:

- Several equity funds delivered strong long-term growth.
- Funds with consistent NAV appreciation recorded higher CAGR values.
- Long-term investments generated better returns compared to short-term holding periods.
- CAGR provided a reliable measure of overall fund performance.

Sharpe Ratio Analysis

The Sharpe Ratio was used to evaluate risk-adjusted returns by comparing fund returns against their volatility. A higher Sharpe Ratio indicates better returns for each unit of risk taken by an investor.

Key Observations:

- Top-performing funds achieved higher Sharpe Ratios.
- Risk-adjusted performance varied across fund categories.
- Some funds generated superior returns with lower volatility.
- Sharpe Ratio helped identify efficient investment opportunities.

Business Insight

CAGR and Sharpe Ratio together provide a comprehensive view of mutual fund performance. While CAGR measures growth potential, the Sharpe Ratio evaluates the quality of returns by considering risk, helping investors make informed investment decisions.

Fund Scorecard Analysis

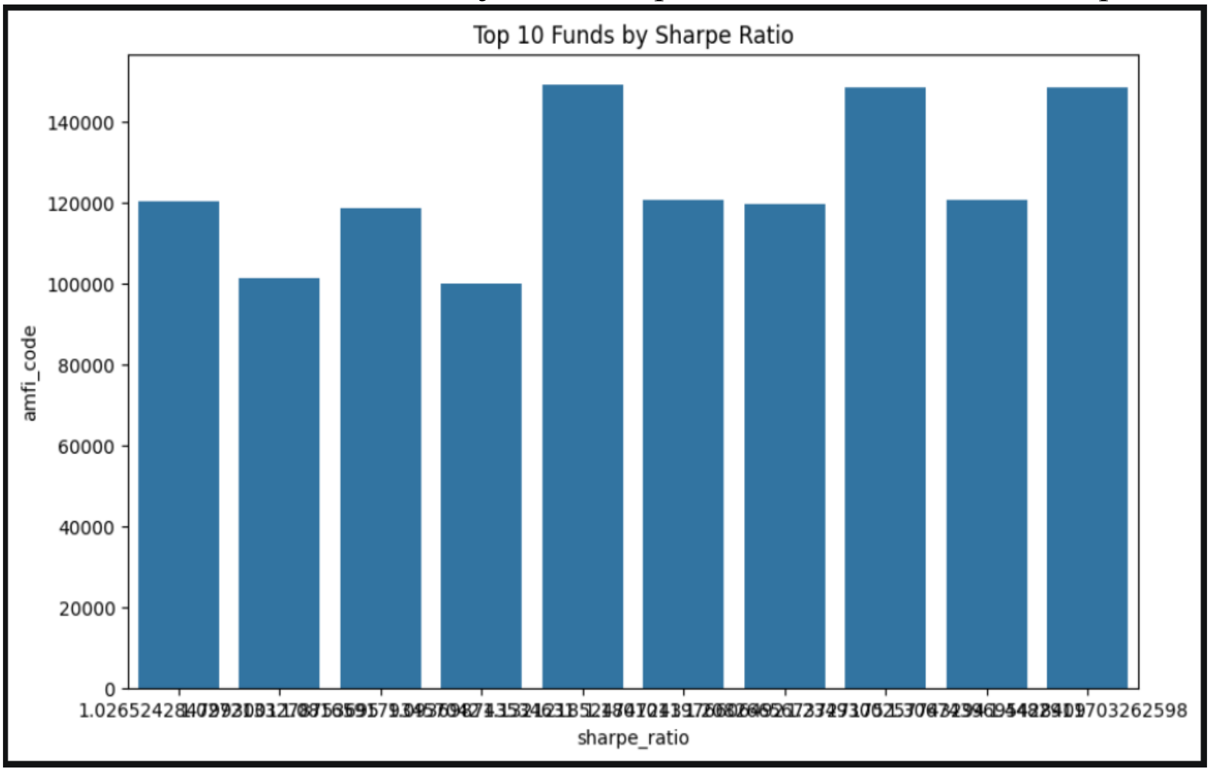
To simplify mutual fund evaluation, a composite Fund Scorecard was developed by combining multiple performance and risk metrics into a single ranking framework. The scorecard considered key factors such as returns, Sharpe Ratio, Alpha, expense ratio, and Maximum Drawdown.

Each fund received a score between 0 and 100, allowing for easy comparison across different schemes. Higher scores indicate stronger overall performance after considering both return potential and risk exposure.

The analysis revealed that funds with consistent returns, higher risk-adjusted performance, and lower drawdowns generally achieved the highest scores. These funds demonstrated the ability to generate stable returns while effectively managing risk.

Key Findings

- Top-ranked funds consistently outperformed peers across multiple metrics.
- High Sharpe Ratio funds achieved better risk-adjusted returns.
- Funds with lower expense ratios benefited from improved overall rankings.
- Lower drawdown funds provided better downside protection during market volatility.
- The scorecard enabled objective comparison between investment options.



Advanced Risk Analytics (VaR & CVaR)

To evaluate downside risk, Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated for all mutual fund schemes. These metrics help estimate potential losses under adverse market conditions and provide a deeper understanding of portfolio risk.
VaR represents the maximum expected loss at a specified confidence level, while CVaR measures the average loss beyond the VaR threshold. Together, these metrics provide a comprehensive assessment of tail risk.

Key Findings

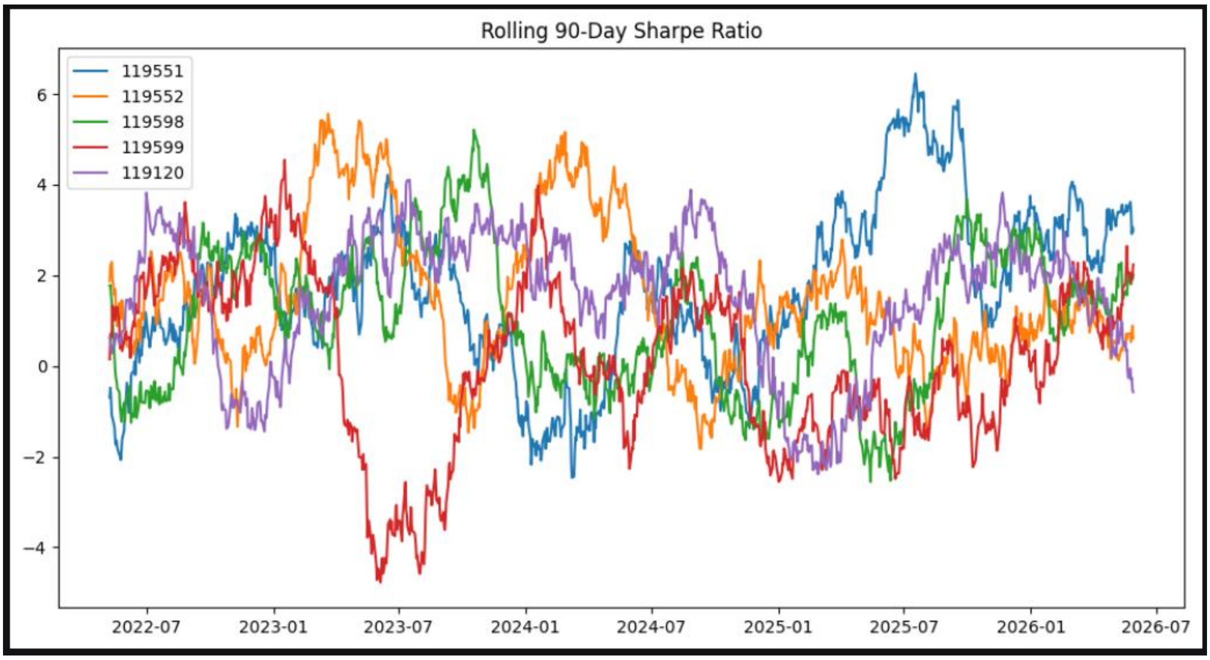
- Funds with higher volatility recorded larger VaR values.
- CVaR values were generally higher than VaR, indicating additional risk during extreme market events.
- Equity-oriented funds exhibited greater downside risk compared to conservative funds.
- Risk levels varied significantly across fund categories.

Rolling Sharpe Ratio Analysis

A Rolling 90-Day Sharpe Ratio was calculated to analyze how risk-adjusted fund performance changed over time. Unlike a static Sharpe Ratio, the rolling approach captures variations in performance across different market conditions. The analysis was conducted for selected mutual fund schemes using a 90-day rolling window. This helped identify periods of strong and weak risk-adjusted returns.

Key Findings

- Risk-adjusted performance fluctuated over time due to changing market conditions.
- Certain funds maintained consistently strong Sharpe Ratios throughout the analysis period.
- Market corrections caused temporary declines in risk-adjusted returns.
- Funds with stable performance exhibited smoother Sharpe Ratio trends.



Investor Analytics

Investor behaviour was analyzed using cohort analysis and SIP continuity analysis. These techniques provide insights into investment patterns, customer engagement, and long-term investment discipline.

The cohort analysis grouped investors based on their first year of investment and measured average investment amounts and overall contributions. SIP continuity analysis evaluated the consistency of recurring investments by calculating transaction gaps.

Key Findings

- Long-term investor cohorts contributed higher total investment amounts.
- SIP investments remained the most popular investment method.
- Most investors maintained regular investment schedules.
- A small group of investors showed irregular investment behaviour and were classified as at-risk.

Fund Recommendation System

A simple recommendation system was developed to assist investors in selecting suitable mutual funds based on their risk appetite. The system categorizes investors into Low, Moderate, and High-risk profiles and recommends top-performing funds using risk-adjusted performance metrics.

The recommendation engine primarily uses Sharpe Ratio rankings and risk grades to identify suitable investment options.

Key Findings

- Moderate-risk funds offered the best balance between return and risk.
- High-risk funds generated higher return potential but also carried greater volatility.
- Low-risk funds provided stability and capital preservation.
- The recommendation engine successfully identified top-performing schemes within each risk category.

Dashboard Overview

An interactive Power BI dashboard was developed to transform analytical results into meaningful visual insights. The dashboard enables users to explore mutual fund performance, investor behavior, and market trends through interactive charts, KPI cards, and filters.

The first dashboard page, **Industry Overview**, provides a high-level summary of the mutual fund industry. Key metrics such as Total AUM, SIP Inflows, Folio Count, and Number of Schemes are displayed using KPI cards. Additional visualizations highlight industry growth trends and fund house performance.

The second dashboard page, **Fund Performance**, focuses on evaluating mutual fund schemes using return and risk metrics. Interactive visualizations allow users to compare funds, analyze scorecards, and review performance trends.

Key Features

- Interactive KPI Cards
- Fund Performance Analysis
- Return vs Risk Comparison
- Fund Scorecard Rankings
- Dynamic Filtering and Slicers
- User-Friendly Navigation

Benefits

The dashboard simplifies complex financial data and enables users to quickly identify high-performing funds, evaluate risk levels, and monitor investment trends through an intuitive visual interface.

Investor Analytics and Market Trends Dashboard

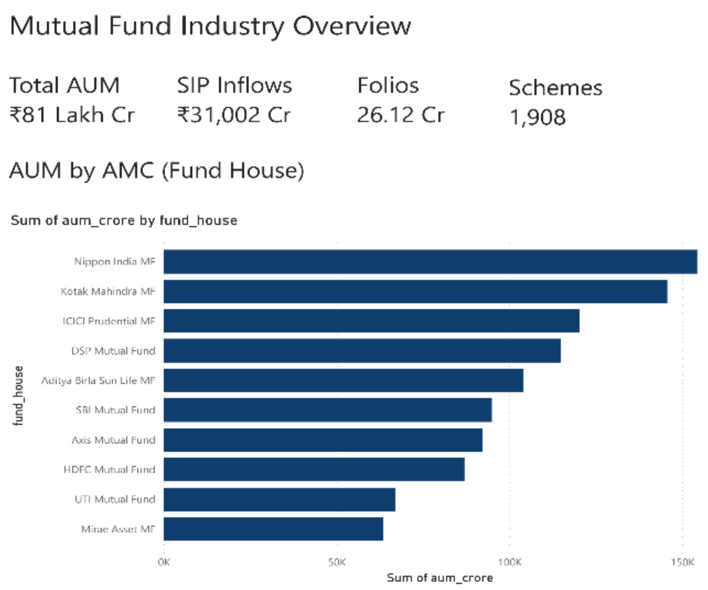
The final dashboard pages focus on investor behavior and investment trends. These dashboards provide valuable insights into transaction patterns, demographic participation, and SIP growth across different regions and investor segments.
The **Investor Analytics** page presents state-wise investment analysis, transaction type distribution, age-group participation, and monthly transaction trends. Interactive filters allow users to analyze investor behavior across multiple dimensions.
The **SIP & Market Trends** page focuses on investment growth patterns and category-level analysis. It highlights SIP trends, investment distribution, and category performance using visual analytics.

Key Findings

- Investor participation increased steadily over time.
- SIP investments remained the dominant investment method.
- Metropolitan regions contributed the highest transaction volumes.
- Category-level analysis revealed differences in investor preferences.

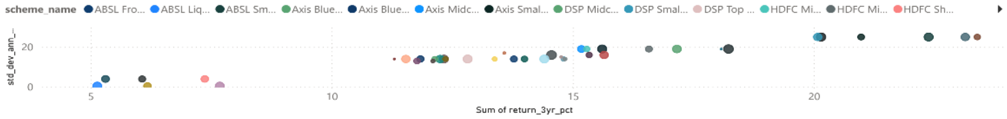
Benefits

These dashboards help stakeholders understand investor behavior, identify growth opportunities, and make data-driven business decisions using visual analytics and interactive reporting tools.



Fund Performance Dashboard

Fund Risk vs Return Analysis



Fund Scorecard Table

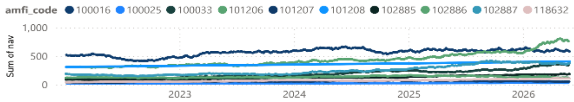
amfi_code	Sum of cagr	Sum of score	Sum of sharpe_ratio	Sum of alpha	Sum of max_drawdown
101208	0.07	22.25	-0.82	0.06	0.00
100025	0.04	17.00	-0.57	0.03	-0.04
118636	0.05	20.63	-0.36	0.03	-0.08
119120	0.06	21.88	-0.23	0.04	-0.04
102886	0.01	28.25	-0.21	-0.12	-0.28
100016	0.03	32.50	-0.20	-0.09	-0.25
120844	0.07	27.88	-0.09	0.07	0.00
Total	**6.69**	**2,050.00**	**21.49**	**0.00**	**-7.15**

fund_house
- (Blank)
- Aditya Birla Sun Life MF
- Axis Mutual Fund
- DSP Mutual Fund
- HDFC Mutual Fund

category
- (Blank)
- ELSS
- Flexi Cap
- Gilt
- Index

plan
- (Blank)
- Direct
- Regular

NAV Trend Analysis



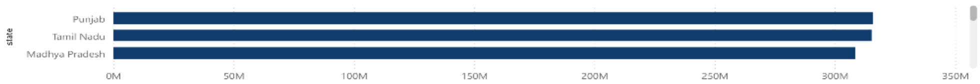
scheme_name
- (Blank)
- ABSL Frontline Equity Fund ...
- ABSL Liquid Fund - Regular ...
- ABSL Small Cap Fund - Reg...

amfi_code

100016 149324

Transaction Amount by State

Sum of amount_inr by state

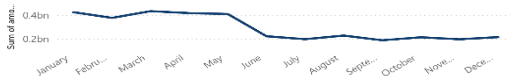


SIP / Lumpsum / Redemption Split

Sum of amount_inr by transaction_type

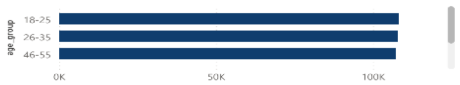
Monthly Transaction Volume

Sum of amount_inr by Month



Age Group vs Average SIP Amount

Average of amount_inr by age_group



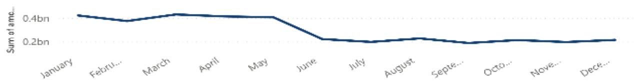
state
- Delhi
- Gujarat
- Haryana
- Karnataka
- Madhya Prad...

age_group
- 18-25
- 26-35
- 36-45
- 46-55
- 56+

city_tier
- B30
- T30

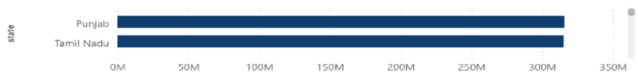
SIP Trend

Sum of amount_inr by Month



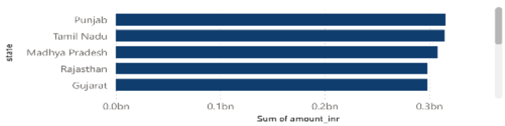
Top Categories by Net Inflow

Sum of amount_inr by state



Category Analysis

Sum of amount_inr by state



Limitations

While the Mutual Fund Analytics project provides valuable insights, several limitations should be considered when interpreting the results.
The project primarily relies on historical datasets and therefore does not fully capture real-time market dynamics. Future market conditions may differ significantly from historical patterns.
Some advanced financial calculations required benchmark assumptions and simplified risk-free rate estimates. These assumptions may affect the accuracy of certain performance metrics.
Additionally, the project uses a limited set of mutual fund schemes and investor records. A larger dataset would improve analytical depth and model reliability.

Key Limitations

- Dependence on historical data.
- Limited benchmark and market datasets.
- Assumed risk-free rate for risk calculations.
- Simplified recommendation logic.
- Limited real-time integration.

Despite these limitations, the project successfully demonstrates practical applications of data engineering, analytics, and business intelligence techniques within the mutual fund domain.

Recommendations and Conclusion

The Mutual Fund Analytics project successfully demonstrates an end-to-end data engineering and analytics workflow, covering data ingestion, validation, transformation, storage, analysis, risk assessment, and dashboard development.
The project provides valuable insights into mutual fund performance, investor behavior, and market trends through advanced analytical techniques and interactive visualizations.

Recommendations

- Integrate live mutual fund APIs for real-time analytics.
- Expand the dataset to include additional fund categories and benchmarks.
- Enhance the recommendation system using machine learning techniques.
- Deploy dashboards online for broader accessibility.
- Implement automated pipeline scheduling and monitoring.

Conclusion

The Mutual Fund Analytics project successfully demonstrated an end-to-end data engineering and analytics workflow, covering data ingestion, validation, transformation, storage, analysis, risk assessment, and dashboard development. The project provided valuable insights into mutual fund performance, investor behavior, and market trends through advanced analytical techniques and interactive visualizations. The developed dashboard and analytical models support data-driven investment decisions and showcase practical applications of data engineering and business intelligence.

GitHub Repository

Project Source Code: GitHub

Author

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Bluestock Internship Capstone Project
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